Project Plan

Radiation-Induced Defect Evolution and Device Degradation in Silicon

Single living roadmap for project architecture, workflow, file map, and module sequencing

Purpose. This document is the single living roadmap for **Radiation Effects Project 2**. The top-level `README.md`, `project_plan.tex`, and `project_plan.pdf` are the maintained architecture documents and should be updated whenever the project architecture changes.

Contents

Current architecture revision. Module 1 is now planned as an **interaction-resolved Geant4 campaign** with separate photon and neutron runs across targeted energy bands so that different microscopic interaction mechanisms can be studied intentionally rather than mixed together in one broad source case. Modules 2–6 are now 2D-first. Module 2 now includes a first executable two-dimensional finite-element Poisson solver using linear triangular elements for defect-dependent space charge. Module 4 is now a single **two-dimensional ballistic-diffusive thermal-transport module** adapted from the ballistic-diffusive heat-conduction framework developed in the FEM_BallisticDiffusion_PINN project. The older Module 4a/4b thermal notes are archived as legacy baseline material. Module 7 remains the scaling and multiscale-prediction module.

1 Project objective

The project builds a physics-based framework for connecting a radiation event in silicon to the eventual degradation of device performance. The goal is not only to model where energy is deposited, but also to understand how that deposited energy creates defects, how those defects evolve with time and temperature, how they alter the internal electric fields and carrier transport, and ultimately how they change the behavior of the device.

The intended physics chain is:

1. incident radiation deposits energy in silicon,
2. deposited energy seeds point defects and effective defect populations,
3. defects migrate, react, cluster, dissociate, and anneal,
4. charged defects alter the internal electrostatic landscape,
5. thermal evolution changes defect kinetics and transport coefficients,
6. the changed field and trap population alter carrier transport,
7. the coupled system produces measurable device degradation.

2 Module roadmap

No.	Module	Purpose
1	Interaction-resolved Geant4 radiation deposition campaign in silicon	Run separate photon and neutron simulations in targeted energy bands so that distinct interaction mechanisms can be isolated, compared, and translated into deposition maps, secondary-particle spectra, and damage-precursor metrics for later modules.
2	Two-dimensional electrostatics with defect-dependent space charge	Solve the 2D electrostatic Poisson problem for a damaged silicon structure using a linear triangular finite-element implementation. Convert dopants and charged defects into electric potential, electric field, and depletion-structure changes.
3	Two-dimensional defect evolution with material-aware kinetic coefficients	Evolve vacancies, interstitials, complexes, and effective defect species in 2D using diffusion, reaction, migration, and annealing laws whose reduced coefficients can depend on temperature, charge state, dopants, and electric field.
4	Two-dimensional ballistic-diffusive thermal transport in silicon	Use a reduced ballistic-diffusive heat-conduction framework to evolve the 2D thermal field in silicon, retain finite-flight-time and boundary-memory effects beyond Fourier conduction, support volumetric radiation-related heat sources, and update temperature-dependent defect and device coefficients. A first executable MATLAB path is now included using a reduced rectangular-domain ballistic-flux closure.
5	Two-dimensional drift-diffusion carrier transport with defect-assisted recombination	Predict how the damaged device transports electrons and holes under the altered field, temperature, and trap landscape.
6	Coupled two-dimensional multiphysics degradation model	Couple Modules 2, 3, 4, and 5 into a self-consistent simulator for defect evolution, field evolution, thermal evolution, and carrier-transport degradation.
7	Multiscale extrapolation and scalable prediction	Develop methods for making predictions on much larger spatial, temporal, or statistical scales than the direct high-fidelity simulation can afford. This module covers controlled approximations, quantified error bars, statistical upscaling, and hybrid multiresolution workflows.

3 Project narrative and module descriptions

Radiation-Induced Defect Evolution and Device Degradation in Silicon

This project builds a physics-based framework for connecting a radiation event in silicon to the eventual degradation of device performance. The goal is not only to model where energy is deposited, but also to understand how that deposited energy creates defects, how those defects evolve with time and temperature, how they alter the internal electric fields and carrier transport, and ultimately how they change the behavior of the device. Each module isolates one layer of the physics, while the final module combines them into a self-consistent multiphysics model.

Module 1. Interaction-Resolved Geant4 Radiation Deposition Campaign in Silicon

This module provides the radiation input to the rest of the framework, but it now does so through a structured set of interaction-targeted simulation campaigns rather than a single broad source run. Using Geant4, the simulation tracks incoming radiation and calculates where, when, and how much energy is deposited inside the silicon while also recording which physical interaction mechanisms dominate in each energy band. The output of this step represents the initial physical insult to the material. In practice, this module gives the spatial and possibly temporal distribution of deposited energy, secondary-particle production, and interaction-specific deposition signatures that can later be translated into an initial defect-generation source term or damage map.

For photons, the current planning assumption is to split the campaign into bands chosen to emphasize Rayleigh scattering, the photoelectric effect, Compton scattering, pair production, photonuclear interactions in the resonance region, and deep-inelastic or very-high-energy behavior. The approximate photon planning bands currently under consideration are eV to about 100 keV for Rayleigh-dominated studies, eV to about 500 keV for photoelectric-dominated studies, roughly 10 keV to 10 MeV for Compton-dominated studies, above 1.022 MeV for pair-production access, roughly 5 MeV to a few GeV for photonuclear studies, and GeV-scale and above for deep-inelastic or high-energy cascade studies. These bands are not interpreted as sharp exclusive boundaries; they are campaign ranges chosen to emphasize different interaction families.

For neutrons, the same philosophy will be used: separate runs will be designed to target different interaction regimes rather than mixing all neutron energies into one undifferentiated source case. The exact neutron energy partition and the dominant reactions to emphasize will be finalized as the Module 1 campaign plan is expanded. In all cases, Module 1 should save not only deposited-energy maps but also interaction tallies, secondary-particle information, and compact diagnostics that explain why one source case produces a different downstream defect precursor than another.

Module 2. Electrostatics with Defect-Dependent Space Charge in a 2D Silicon Device Cross Section

This module models how radiation-induced defects modify the electrostatic state of the device in two spatial dimensions. Defects can act as charged centers, traps, or ionized states that contribute to the net space-charge density. That altered space charge changes the electric potential and electric field inside the device. Physically, this matters because the internal field controls depletion behavior, carrier motion, and ultimately how the device responds under bias. This module therefore connects the evolving defect population to the device's internal field structure.

The current implementation solves the scalar Poisson equation with a Galerkin finite-element method on a rectangular triangular mesh. The nodal unknown is electrostatic potential, the element matrix is assembled from $\int \varepsilon_{\mathrm{Si}} \nabla N_a \cdot \nabla N_b \, d\Omega$, the source vector is assembled from $\int N_a \rho \, d\Omega$, Dirichlet contact potentials are imposed strongly, and homogeneous Neumann boundaries enter naturally. The first verification cases check the zero-charge limit, linear Laplace solution, uniform space-charge limit, and localized charged-defect perturbation.

Module 3. Two-Dimensional Diffusion-Reaction Solver for Defect Migration and Annealing

This module describes the time evolution of radiation-induced defects after they are created. Defects are not static: they can diffuse, recombine, cluster, dissociate, or anneal depending on temperature and local conditions. In the upgraded Module 3 plan, the diffusion and annealing coefficients are treated as material-aware reduced kinetic coefficients rather than as universal constants. Their microscopic origin lies in bonding, lattice geometry, and electronic structure, while the first implemented simulation-level dependencies are temperature, defect charge state, dopant environment, and electric field. A 2D diffusion-reaction model then evolves the resulting defect concentrations in space and time. This is one of the core physics modules of the project because it determines how initial damage develops into a changing defect landscape, rather than treating radiation damage as frozen in place.

Module 4. Two-Dimensional Ballistic-Diffusive Thermal Transport in Silicon

This module adds temperature as an active physical driver using a reduced ballistic-diffusive heat-conduction model derived from the carrier Boltzmann picture. The purpose is to retain a clearer microscopic foundation than a plain Fourier heat equation while still keeping the solver executable on the same 2D device cross section used by the other modules. The thermal field controls defect diffusivities, annealing rates, and later carrier-transport coefficients. In the revised architecture, Module 4 accepts volumetric source terms appropriate for radiation-related heating and retains an explicit ballistic contribution through a first reduced rectangular-domain boundary-emission closure so that finite-flight-time and boundary-memory effects can already be represented in the initial implementation.

Module 5. Two-Dimensional Drift-Diffusion Carrier Transport with Defect-Assisted Recombination

This module translates material damage into electrical performance changes. Using 2D drift-diffusion transport equations, the model solves for electron and hole motion under the influence of the internal electric field and carrier concentration gradients. Radiation-induced defects enter through trap-assisted recombination, trapping, and lifetime degradation. In other words, this module shows how the evolving defect population, altered electrostatics, and temperature field affect measurable device behavior such as leakage current, charge collection, recombination losses, or degraded diode response.

Module 6. Combined Multiphysics Model Using Modules 2, 3, 4, and 5

This module integrates the previous physics pieces into a coupled framework. The defect population changes the space charge; the space charge changes the electric field; the electric field influences carrier transport; the temperature affects defect migration and annealing; and carrier behavior and local power dissipation can in turn interact with the thermal state. This module is where the project becomes a true device-degradation simulator rather than a collection of separate models. Its purpose is to produce a self-consistent prediction of how radiation exposure evolves into long-term electrical degradation.

Module 7. Multiscale Extrapolation and Scalable Prediction

This module addresses the fact that direct high-fidelity simulations may be too expensive for large devices, long times, broad parameter sweeps, or population-level reliability predictions. It covers controlled reduced-fidelity approximations, statistical upscaling, and hybrid multiresolution workflows. A central requirement is that any speedup or approximation be accompanied by a quantitative error metric relative to a higher-fidelity reference.

4 Module 3 kinetic-coefficient hierarchy

Module 3 now treats the defect diffusivity D and annealing coefficient k_{ann} as reduced constitutive quantities whose values are shaped by underlying material physics. The project explicitly separates two levels of description.

At the microscopic level, defect motion and defect removal are controlled by interatomic bonding, local lattice geometry, and electronic structure. Those factors determine migration barriers, reaction barriers, jump distances, charge-state energetics, and attempt frequencies. They are the physical origin of the Arrhenius-type activation energies and prefactors used in the continuum model.

At the simulation level, Module 3 promotes the coefficients from simple functions of temperature to state-dependent reduced laws. The first implemented dependence set is

$$D = D\left(T, q_d, N_{\text{dop}}, |\mathbf{E}|\right), \tag{4.1}$$

and

$$k_{\text{ann}} = k_{\text{ann}}\left(T, q_d, N_{\text{dop}}, |\mathbf{E}|\right), \tag{4.2}$$

where q_d is the defect charge state, N_{dop} is a dopant-environment measure, and $\mathbf{E}$ is the device-scale electric field imported from Module 2 when that coupling is enabled. Strain and interface proximity are planned as the next refinement layer, while bonding and electronic structure remain the microscopic origin of the reduced parameters rather than direct continuum fields.

This hierarchy keeps the project executable while still making clear which physical effects actually contribute to the effective kinetic coefficients.

5 Module 1 interaction-resolved campaign plan

Module 1 is now organized as an interaction-resolved Geant4 campaign rather than a single undifferentiated radiation-deposition run. The purpose is to select source energies deliberately so that different microscopic interaction mechanisms can be emphasized, compared, and interpreted in downstream damage modeling.

5.1 Why this change matters

Different interaction mechanisms can produce very different spatial deposition patterns, secondary-particle populations, and damage precursors even when the total incident energy is similar. Separating the runs by targeted interaction regime makes it easier to explain why one source case leads to dense local energy deposition, another to broader electron-mediated energy transfer, and another to recoil nuclei or higher-energy secondaries that may matter more for defect production.

5.2 Photon campaign structure

Interaction emphasis	Approximate planning range	Reason for separate campaign
Rayleigh scattering	eV to about 100 keV	Isolate elastic scattering and angular redistribution with relatively small direct energy transfer per event.
Photoelectric effect	eV to about 500 keV	Study strong local energy absorption and dense short-range secondary-electron production.
Compton scattering	about 10 keV to about 10 MeV	Study partial-energy transfer, recoil-electron generation, and broader distributed deposition.
Pair production	above 1.022 MeV	Study electron-positron creation and more extended shower-like deposition behavior.

Photonuclear or resonance-region interactions	about 5 MeV to about 3 GeV	Study photonuclear reactions, recoil nuclei, and higher-energy secondaries that can change defect-precursor character.
Deep-inelastic or very-high-energy interactions	GeV scale and above	Study high-energy cascades and hadronic secondary production in the extreme-energy regime.

These are campaign ranges rather than sharp exclusive boundaries. Multiple processes can contribute in overlapping regions, but the ranges are still useful because they bias the simulation toward distinct physical mechanisms and signatures.

5.3 Neutron campaign structure

Neutrons should be treated with the same philosophy: use separate campaign families to emphasize different reaction classes and energy regimes rather than combining all neutron energies into one broad source case. The exact neutron bands are left as a planned design task so that they can be chosen deliberately around the reactions that matter most for silicon damage, displacement production, recoil spectra, and secondary charged-particle generation.

5.4 Recommended Module 1 outputs

Each interaction-resolved Module 1 run should save, at minimum, a deposited-energy map, depth and lateral deposition summaries, interaction-type tallies, secondary-particle spectra or counts when available, and metadata describing the incident particle, targeted energy band, physics list, geometry revision, binning, and units. This turns Module 1 into both a source generator and an interpretation layer for downstream defect modeling.

6 Core architecture decision: 2D for Modules 2–6

The original one-dimensional path was numerically attractive, but it suppresses several effects that become important once the damage pattern is spatially structured: lateral spreading of defect populations, lateral field bending, nonuniform temperature fields near localized energy deposition, and current crowding or carrier-flow rerouting. A 2D architecture is therefore the more faithful bridge between a realistic Geant4 source pattern and later electrostatic, thermal, and transport calculations.

This move to 2D does not mean every solver should start maximally complex. The first versions should still keep simple rectangular domains, structured grids, small defect networks, simple boundary conditions, and weak one-way coupling before self-consistent feedback.

7 Run-order workflow for the project

The run order below is the logical flow of a coupled study.

Step 1. Module 1: Geant4 radiation transport
Run the Geant4 silicon geometry and score deposited energy, hit coordinates, optional timing information, and any derived damage-source products.

Step 2. Geant4-to-MATLAB handoff
Bin or interpolate the Geant4 output into a 2D deposition map or initial defect-source map on a MATLAB-side grid.

Step 3. Module 3: defect evolution
Evolve the 2D defect state under diffusion, reactions, clustering, dissociation, annealing, and optional source terms.

Step 4. Module 4: ballistic-diffusive thermal field
Solve or update the 2D temperature field with the reduced ballistic-diffusive thermal model and use it to modify defect and transport coefficients.

Step 5. Module 2: electrostatics
Build the net space-charge density from dopants and charged defects, then solve for electric potential and electric field.

Step 6. Module 5: carrier transport
Solve the 2D electron and hole transport problem with field-driven drift, concentration-driven diffusion, and defect-assisted recombination or trapping.

Step 7. Module 6: coupled iteration
Iterate Modules 2, 3, 4, and 5 until the defect state, temperature field, electric field, and carrier state are mutually consistent to a chosen tolerance.

Step 8. Module 7: scalable prediction
Apply reduced-order, approximation, statistical-upscaling, or hybrid multiresolution methods to extend the prediction to larger devices, longer times, larger domains, or rarer-event statistics.

Important distinction. The revised Module 4 is not a plain Fourier baseline. It is a reduced ballistic-diffusive thermal model chosen because it preserves more of the microscopic transport physics while remaining executable in 2D. In simplified studies, thermal information does not always have to be solved before electrostatics, but in the intended coupled workflow the temperature field should be available before the tightly coupled 2D solve because temperature changes diffusion, annealing, mobility, and recombination parameters.

8 Recommended coding order

1. **Module 1:** Geant4 geometry, scoring, and export format.
2. **Module 3:** 2D defect evolution on a structured grid.
3. **Module 2:** 2D electrostatics with prescribed charged defects; first FEM Poisson path now implemented and ready for coupling to Module 3 defect fields.
4. **Module 4:** 2D ballistic-diffusive thermal solver plus temperature-dependent coefficient handoff to later modules.
5. **Module 5:** 2D drift-diffusion with prescribed electrostatic field, thermal field, and defect/trap fields.
6. **Module 6:** weak coupling, then stronger feedback and iterative convergence.
7. **Module 7:** scalable-prediction methods after at least one credible high-fidelity coupled workflow exists.

9 File-map overview

Path	Role
`README.md`	Public repository summary and high-level entry point.
`project_plan.tex/.pdf`	Single living roadmap document: project objective, module definitions, run order, file map, interfaces, and roadmap updates.
`docs/physics_notes/`	One large physics note per module, with derivations, symbol definitions, assumptions, and physical interpretation.
`docs/implementation_notes/`	Module-by-module implementation specifications, solver notes, and interface notes.
`geant4/`	Geant4-side setup, interaction-resolved campaign macros, and example outputs for Module 1.

`matlab/`	MATLAB drivers, source code, tests, outputs, and cross-module interfaces for Modules 2–7.

9.1 Current MATLAB map plus planned 2D emphasis

Path	Role in the updated 2D plan
`matlab/main_module2_electrostatics.m`	Top-level entry for the 2D electrostatic solve.
`matlab/main_module3_2d_defect_evolution.m`	Top-level entry for the 2D defect-evolution solve.
`matlab/main_module4_2d_ballistic_diffusive_thermal.m`	Top-level entry for the Module 4 two-dimensional ballistic-diffusive thermal solve; first executable MATLAB implementation included.
`matlab/main_module5_drift_diffusion.m`	Top-level entry for the 2D carrier-transport solve.
`matlab/main_module6_multiphysics.m`	Top-level coupled driver that orchestrates Modules 2, 3, 4, and 5.
`matlab/src/grid/`	Shared 2D grid generation and spatial operators.
`matlab/src/module3_defects/`	Module 3 defect evolution operators and initialization helpers.
`matlab/src/module4_thermal/`	Module 4 ballistic-diffusive thermal operators, initialization helpers, source-term assembly, reduced ballistic-flux closures, and verification support.
`matlab/src/module2_electrostatics/`	Implemented Module 2 finite-element Poisson operators, triangular mesh generation, charge assembly, boundary-condition insertion, and field postprocessing.
`matlab/src/transport/`	Planned drift-diffusion operators, recombination models, mobility models, and current extraction.
`matlab/src/diagnostics/`	Shared diagnostics, error metrics, and summary writers.
`matlab/src/io/`	Shared I/O helpers and history save utilities.
`matlab/src/plot/`	Shared 2D visualization and history plotting utilities.
`matlab/tests/`	Verification problems and regression tests, including Module 2 electrostatic limiting-case checks.
`matlab/outputs/`	Saved run products, plots, summaries, and histories.

10 Thermal-model rationale and comparison strategy

10.1 Why the ballistic-diffusive model is now primary

The revised Module 4 becomes the primary thermal branch because it preserves a clearer microscopic connection to finite-speed heat-carrier transport than the old continuum baseline while still remaining executable in 2D. The model is more ambitious than Fourier conduction, but it is still substantially simpler than a full angularly resolved phonon Boltzmann solve.

10.2 How it should still be benchmarked

Even though the split 4a/4b architecture is removed, it is still useful to compare the new Module 4 against a simple Fourier benchmark when validating the implementation. Useful metrics include

- peak temperature and hot-spot width,
- early-time surface or boundary heat flux,
- lateral spreading rates of a localized thermal pulse,
- annealing-rate differences induced through temperature changes,
- downstream defect-history changes after thermal feedback is coupled back into Module 3,
- added computational cost.

A useful rule is that the ballistic-diffusive module should earn its additional complexity by changing one or more project-level quantities of interest enough to matter.

11 Module interfaces in the new 2D plan

11.1 Module 1 to Module 3 handoff

The first strong interface should now be a 2D deposition or defect-source map accompanied by interaction-resolved metadata. A minimal handoff object should contain coordinate arrays, a deposited-energy density field, optional time-resolved source information, interaction-type tallies or summaries, secondary-particle diagnostics when available, and metadata describing the incident particle, targeted energy band, device geometry, physics list, bin width, radiation case, and units.

11.2 Module 3 to Module 2 handoff

After a defect-evolution run, MATLAB should save at minimum coordinate arrays, effective defect fields, charged-defect density fields, trap-density fields, and time-history diagnostics for provenance.

11.3 Module 2 and Module 4 to Module 5 handoff

Carrier transport should read electric potential $\phi(x, y)$, electric-field components $E_x(x, y)$ and $E_y(x, y)$, temperature $T(x, y)$, and trap or recombination parameters derived from the defect state.

12 Verification strategy under the 2D architecture

Each module needs at least one low-complexity verification path.

Module	Verification logic
1	Compare deposition trends against known silicon behavior and verify that the exported 2D binning is internally consistent.
2	Use manufactured or symmetric charge configurations whose 2D potential and field patterns are physically obvious; verify sign conventions and boundary-condition behavior.
3	Check pure diffusion, pure annealing, and symmetry-preserving source cases in 2D. Total inventory should behave correctly under the chosen boundaries.
4	Recover expected limits of the 2D ballistic-diffusive thermal model, including equilibrium preservation, localized-pulse spreading, and measurable deviations from a Fourier baseline when boundary-memory effects matter; formal diffusive-limit comparison remains the next validation step.
5	Verify defect-free transport first, then turn on simple trap-assisted recombination. Check that zero-field or symmetric cases behave sensibly.
6	Confirm that decoupled or weakly coupled limits reduce to standalone solutions. Track iterative residuals and relaxation behavior.
7	Every scalable-prediction method must be benchmarked against a higher-fidelity reference. The benchmark should report both speedup and error metrics.

13 Module 7: multiscale extrapolation and scalable prediction

A stronger title than "scaling up methods of the simulation" is **Multiscale Extrapolation and Scalable Prediction**. This module has three important branches:

1. **Controlled reduced-fidelity approximation:** coarser meshes, simplified defect networks, operator splitting, or reduced-order models. These must be judged by both speedup and error relative to a higher-fidelity reference.

2. **Statistical approximation and upscaling:** using smaller simulated tiles, event statistics, or surrogate source distributions to infer larger-scale behavior.

3. **Hybrid multiresolution coupling:** high fidelity only in physically critical regions, with coarser or cheaper models elsewhere.

A useful project-level trade plot is approximation error versus speedup rather than speed alone.

14 When this project-plan document should be updated

This document should be revised whenever a module definition changes, the dimensionality assumption changes, a new top-level file or folder becomes part of the standard workflow, an inter-module file format changes, the run order changes, a major verification strategy changes, or Module 7 gains a concrete benchmark or scaling method.

15 Near-term next steps after this revision

1. Keep the top-level `README.md`, `project_plan.tex`, and `project_plan.pdf` synchronized whenever the project architecture changes.
2. Verify the first 2D Module 3 code path on the user's MATLAB environment.
3. Completed: implemented the first 2D reduced ballistic-diffusive Module 4 MATLAB path with uniform-equilibrium, localized-pulse, and boundary-heating cases.
4. Keep the Module 4 note, file map, and run-order documentation synchronized with the implemented single-module thermal architecture.
5. Update the Module 2, 4, and 5 implementation notes so their first solver architecture is 2D rather than 1D where applicable.
6. Decide the first benchmark quantity of interest for Module 7 so that future approximations can be measured against something physically meaningful.